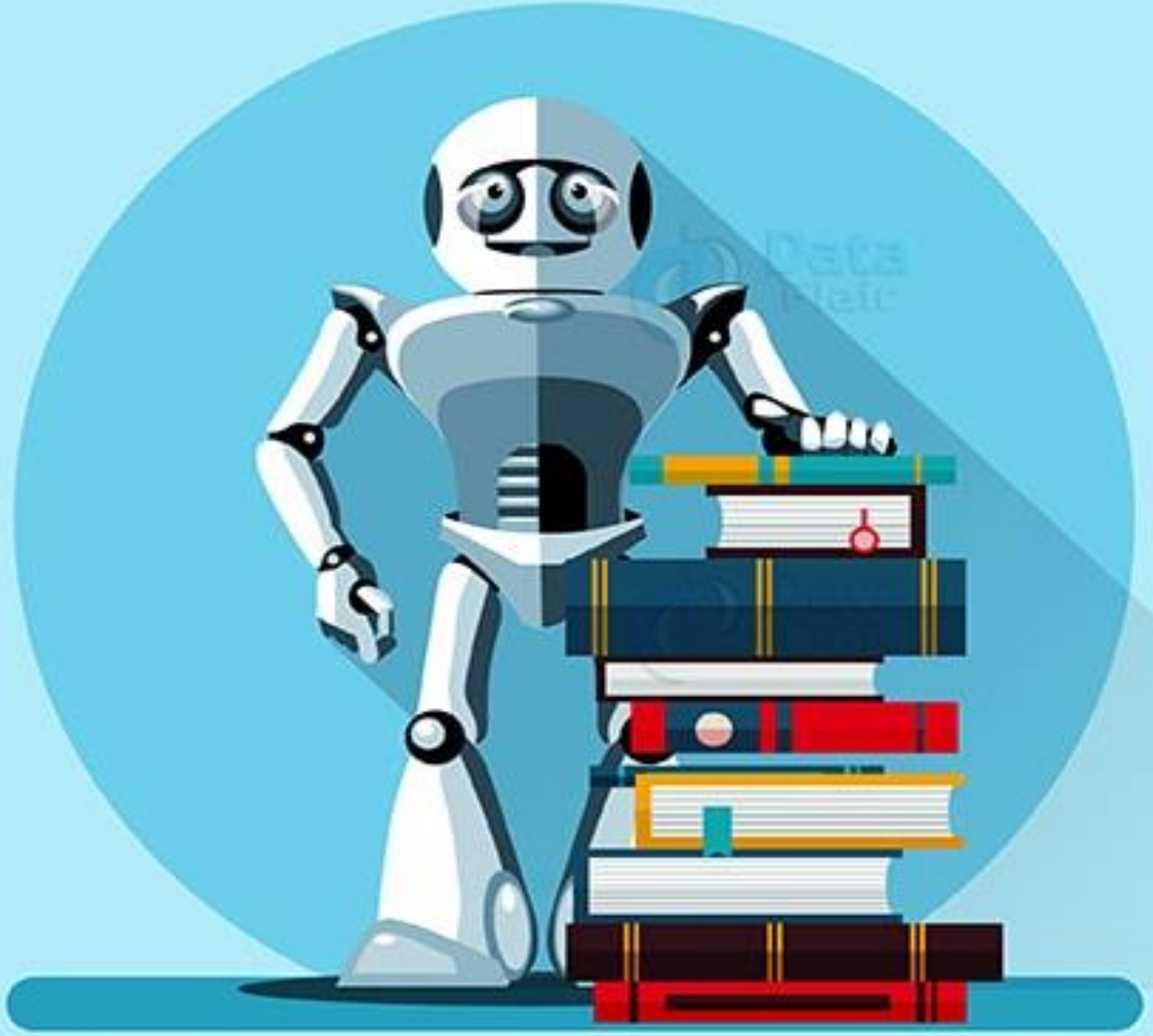
Introduction to Machine Learning(ML)

Introduction to Machine learning



What is machine learning?

Machine learning (ML) is an algorithm that allows software applications to predict the outcomes accurately without being explicitly programmed. The basic function of machine learning is to build algorithms that can receive input data and use statistical analysis to predict an output.

Traditional Programming

Machine Learning

How machine learning works?

- Machine learning algorithms are of 2 types- supervised and unsupervised.
- Supervised algorithms require a data analyst with the required skills to provide both input and desired output.
- Unsupervised algorithms use an iterative approach called deep learning to review data and arrive at conclusions. These algorithms are used for more complex processing tasks than supervised learning systems like image recognition, speech-to-text and natural language generation.

Examples of machine learning

- Siri & Cortana
- Facebook
- Google maps
- Google search
- Netflix
- Uber
- Gmail

Key elements of machine learning

- **Representation:** The way knowledge is represented. Like including decision trees, sets of rules, instances, graphical models, neural networks, support vector machines, model ensembles, etc.
- **Evaluation:** The way programs are evaluated. For example accuracy, prediction and recall, squared error, likelihood, posterior probability, cost, margin, entropy k-L divergence, etc.
- **Optimization:** The way to generate candidate programs known as the search process. For example combinatorial optimization, convex optimization, constrained optimization.

Steps of machine learning

- **Collecting data:**

Data can be written on paper, recorded text files and spreadsheets or stored in an SQL database. It is to be stored in an electronic format suitable for analysis.

- **Preparing the data:**

The quality of a machine learning project is based on the quality of the data it uses. It is suggested that 80% of the effort is put into data preparation.

- **Training the model on the data:**

The specific machine learning task will inform the selection of an appropriate algorithm. Then “feed” the data to the model during this phase and get a learner. A learner is a ML algorithm that has been trained on some data and adjusted to fit the same.

- **Evaluating model performance:**

As each learner results in a biased solution, it is important to evaluate how well the algorithm learned from its experience. Depending on the model used, evaluation of the accuracy of the learner using a test dataset might be possible.

- **Improving model performance:**

If better performance is needed, it is necessary to use more advanced strategies to improve the model's performance, or switch to a different model, add additional data and perform additional work on the data. After this, if the model appears to be performing properly, it can be used for its original task.

- **Prediction:**

Machine learning is using the data to answer questions. This is the point where the value of machine learning is realized and finally the model can be used to predict the outcome.

Advantages of Machine learning

- **Data input from unlimited sources**
  It can collect large data with timely analysis and assessment. It helps to review and adjusts your message based on recent customer interactions and behaviours and also pinpoints relevant variables.

- **Fast Processing and real time predictions**
  Machine learning algorithms operate at expedited levels and due to it's speed in consuming data it is able to tap burgeoning trends and produce real time predictions.

- **Practical scenarios** This use is vital. Predictive analytics as well as their impact on real life situations pertaining to customer acquisitions is important to understand.

- **Machine learning in medical industries**

It is used to predict admission rates. Medical systems incorporate these technologies to streamline and centralize expense reports and testing protocols.

- **Machine Learning in insurance industry**

It predicts different types of insurance and coverage plans; predicts policy updates; predicts fraudulent insurance claim volumes and also establishes solutions based on artificial intelligence.

Disadvantages of Machine Learning

- **Error diagnosis and correction**
  Actual problem is when the do make errors, it is really difficult to diagnose and correct them because it will require going through underlying complexities of algorithms.
- **Time constraints in Learning**
  It is quite impossible to make accurate predictions with a machine learning system because it learns through historical data. So, bigger the data, longer the exposure to data, better the performance.
- **Problems with verification**
  It lacks variability. It deals with statistical truths not literal truth. Certainty of the situations excluded from historical data will be difficult to prove.
- **Limitations of predictions**

They don’t always provide a rational reason for a prediction.

It is basically prone to hidden and unintentional biases . Machine learning systems are to answer questions , not to pose them in which human collaboration is somewhat important.

Machine learning *is an enabler for analysis of massive quantities of data. While it delivers faster, more accurate results to identify profitable opportunities or dangerous risks, it may also require additional time / resources to train it properly.*

Types of Machine Learning At A Glance

Supervised Learning

- Makes machine Learn explicitly
- Data with clearly defined output is given
- Direct feedback is given
- Predicts outcome/future
- Resolves classification and regression problems

Unsupervised Learning

- Machine understands the data (Identifies patterns/structures)
- Evaluation is qualitative or indirect
- Does not predict/find anything specific

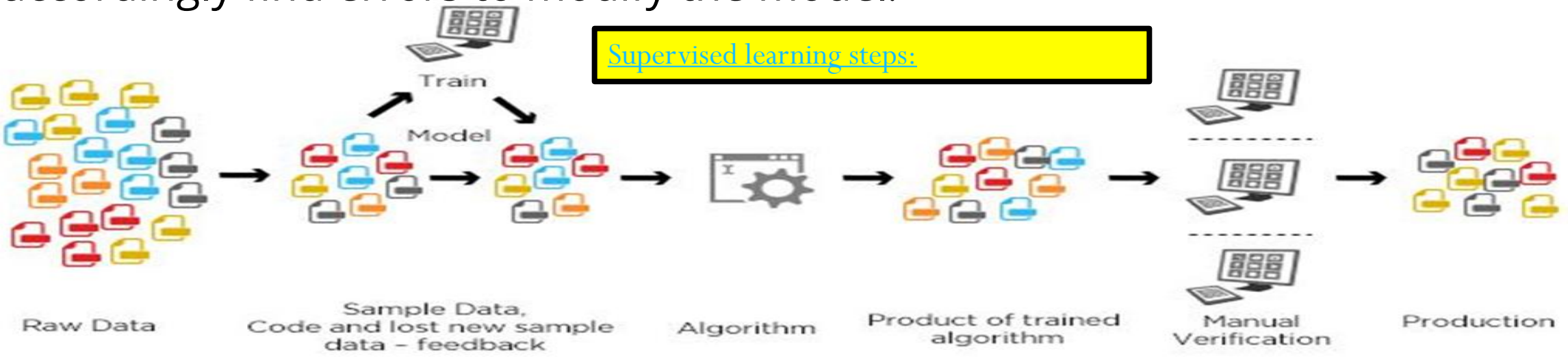
Reinforcement Learning

- An approach to AI
- Reward based learning
- Learning form +ve & +ve reinforcement
- Machine Learns how to act in a certain environment
- To maximize rewards

Supervised machine learning

- **Supervised machine learning algorithms** can apply basis past learnings to new data using examples labeled to predict future events. With start of analysis of a known training dataset, the learning algorithm produces an inferred function and make predictions about the output values. After sufficient training, system is able to provide targets for new input. The learning algorithm can compare output with the correct and intended output and accordingly find errors to modify the model.

Supervised learning steps:



Supervised machine learning

Supervised Learning

BASIS FOR COMPARISON	CLASSIFICATION	REGRESSION
Basic	The discovery of model or functions where the mapping of objects is done into predefined classes.	A devised model in which the mapping of objects is done into values. Ex:- House Price Prediction, where the price of the house will predict from the inputs such as No of rooms, Locality, Ease of transport, Age of house, Area of a home.
Involves prediction of	Discrete values	Continuous values
Algorithms	Decision tree, logistic regression, etc.	Regression tree (Random forest), Linear regression, etc.
Nature of the predicted data	Unordered	Ordered
Method of calculation	Measuring accuracy	Measurement of root mean square error

Types of Classification Algorithms

Some of the main classification algorithms that are present in machine learning and used for different classification applications are as follows-

Logistic Classification:-

Under the category of supervised learning; Logistic regression measures the relationship between the dependent variable (Binary or Dichotomous with only two possible values like "Yes or No", "Living or Dead") which is categorical with one or more than one independent variables by estimating probabilities using a logistic function.

K-Nearest Neighbors:-

KNN algorithm is one of the most used and straightforward learning algorithms in classification. Major vote of an object is classified by its neighbors, with purpose assigned to the class most common among its k nearest neighbors .

Kernel Support Vector Machines:-

Kernel-SVM algorithm is mathematical functions set. The purpose of core is to take input data and transform that into required form. Different SVM algorithms use different types of kernel functions e.g. linear and nonlinear functions, polynomial functions, radial basis function, and sigmoid functions.

Decision Tree Classification:-

Decision tree classification have models in the form of a tree structure that can handle both categorical and numerical data. An associated decision tree incrementally developed and simultaneously It breaks down large data-set into smaller subsets. The final outcome is a tree with decision nodes and leaf nodes.

Types of Regression Algorithms

Some of the main Regression algorithms that are present in machine learning and used for different Regression applications are as follows-

Simple Linear Regression:-

In simple linear regression, score prediction takes place on one variable from the ratings on a second variable. The variable forecasting is called the criterion variable and referred to as Y. The variable based on predictions is called the predictor variable and denoted to as X.

Multiple Linear Regression:-

Multiple linear regression technique is the most common form of linear regression analysis. As a predictive analysis, the multiple linear regression is used to define the relationship between one dependent variable with two or more than two independent variables which can be continuous or categorical.

Polynomial Regression:-

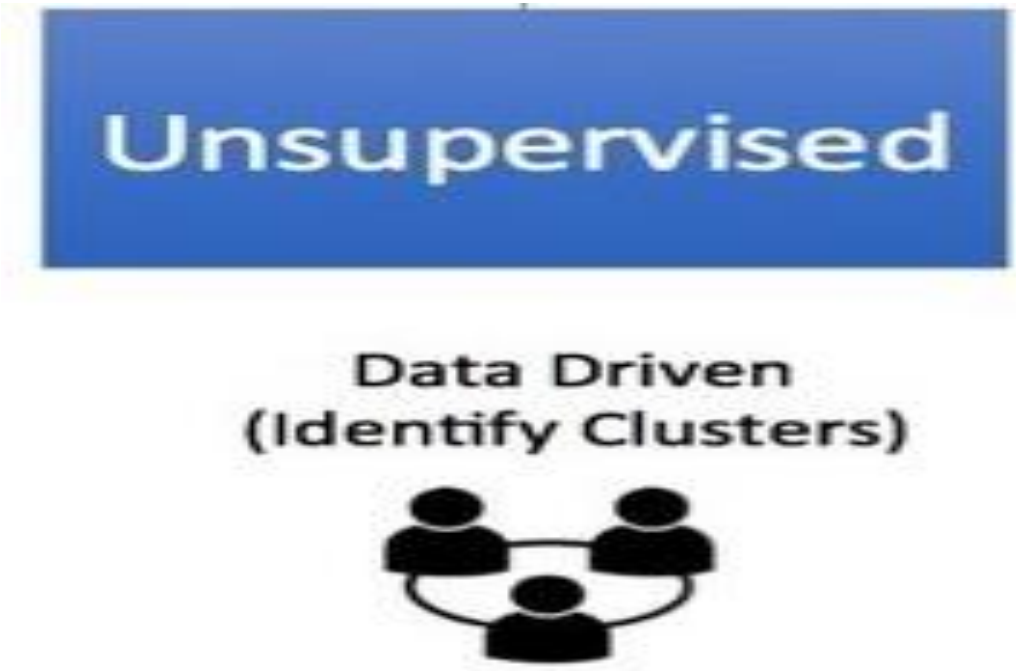
Polynomial regression is another form in which maximum power of the independent variable is more than 1 and best fit line is not a straight line instead it is in the form of a curve.

Ridge Regression:-

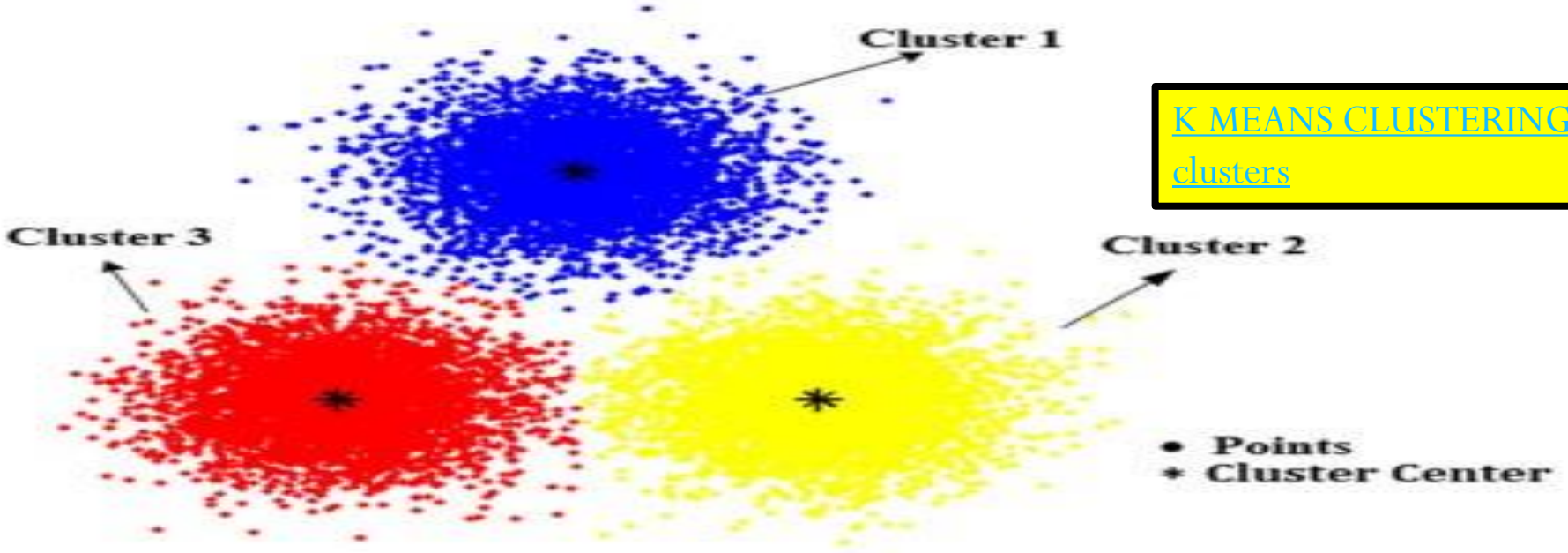
Ridge Regression is one of the algorithms used for analysing multiple regression data that suffer from multi collinearity. By addition of a degree of bias to regression calculates, it reduces the standard errors and the net effect will be to give calculations that are more reliable.

Unsupervised machine learning

- **Unsupervised machine learning algorithms** are used wherein information used to train is neither classified nor labeled. Unsupervised learning studies establishes how systems can infer a function to describe a hidden structure from unlabeled data.



Types of Unsupervised Learning



K MEANS CLUSTERING showing 3 clusters

K-Means Clustering:-

K-Means clustering where similar data grouped in a cluster and It starts with K as the input centroids in random locations in the space. With the use of the **Euclidean distance method** calculate the distance between data points and centroids, and assign data point to the cluster which is close to it. Recalculate the cluster centres as a mean of data points attached to it and repeat until no further changes occur.

Hierarchical Clustering:-

Hierarchical clustering have similar data grouped in a cluster. It is an algorithm that builds the hierarchy of clusters. The algorithm starts with all the data points assigned to a bunch of their own and then two nearest groups are merged into the same cluster. In the end, this algorithm terminates when there is only a single cluster left.

Principal Component Analysis:-

Principal Component Analysis is an algorithms of Dimensionality Reduction, in this technique, it is transformed into a new set of variables from old variables, which are the linear combination of real variables. Specific new set of variables are known as principal components and as a result of the transformation, the first primary component has the most significant possible variance, and each following element has the highest potential difference under the constraint that it is orthogonal to the above ingredients.

Linear Discriminant Analysis:-

The linear discriminant analysis is one of the algorithms of Dimensionality Reduction in which it also creates linear combinations of original features and it optimizes the reparability between classes. LDA can improve the predictive performance of the extracted features and offers variations to tackle specific roadblocks.

Reinforcement machine learning

- **Reinforcement machine learning algorithms** interacts with its environment by producing actions and discovers errors or rewards. Trial and error search and delayed reward are most relevant characteristics of reinforcement learning.

Element of reinforcement learning

- **Agent**: Intelligent programs
- **Environment**: External condition
- **Policy**:
  - Defines the agent's behavior at a given time
  - A mapping from states to actions
  - Lookup tables or simple function

Environment: Physical world in which the agent operates

State: Current situation of the agent

Reward: Feedback from the environment

Policy: Method to map agent's state to actions

Value: Future reward that an agent would receive by taking an action in a particular state

Types of Reinforcement Learning

Reinforcement Learning algorithms are present in machine learning, which applied for different learning applications and some of the main algorithms are as follows-

Q-Learning:-

Q-learning is an agent attempts to learn the optimal strategy from its history of communication with the environment. Q-learning learns an optimal policy no matter which procedure the agent is following as long as there is no restriction on the plenty of times it tries an action in any state.

Markov Decision Processes:-

Markov Decision Process is a straightforward framing of the problem of learning from interaction. The agent was selecting actions and the environment responding to these actions, and the agent and the environment interact continually and presenting new situations to the agent.

MACHINE LEARNING, DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

› All these 4 are inter-related to each other.

Artificial Intelligence

- **When computer has abilities of imitating aspects of human intelligence. Such areas include pattern recognition, vision and reasoning .[1]**
  Example: Google home
- AI imitates human senses like hearing, vision etc.

Machine Learning

- **Branch of AI based on the idea that the systems can learn from data, identify patterns and make decisions with minimal human interventions.[2]**
- Applications include spam detection, sales forecasting, product recommendation- based on the previously obtained similar data.

Deep Learning

- **Specialized part of ML having artificial neural networks. It is when ML is trying to imitate human brain.**
- Applications include language transactions, speech and language recognition, etc.

Data Science

An interdisciplinary field of system and process to obtain information from data in various forms. It builds and modifies AI software to obtain information from large amount of data/ clusters of data[3]

Source 1 : medium.com/@edd/bluffers-guide-to-ai-deep-learning-and-data-science-da350a6f9b3
Source 2 : sas.com/en_in/insights/analytics/machine-learning.html
Source 3 : greatlearning.in/blog/difference-data-science-machine-learning-ai/

INDUSTRIAL APPLICATIONS OF MACHINE LEARNING

- Machine learning has become a part and parcel of our daily works. Online shopping on amazon and other domains, using google maps for Traffic route, use of online social platforms to connect with friends (Facebook) are all examples where machine learning has been helpful.

- In the current era, almost every industry requires Machine Learning to survive and maintain it's position in the ever-growing competitive world where it is necessary to find efficient ways and gain advantage over the competitors.

USE OF MEACHINE LEARNING IN DIFFERENT SECTORS

ML IN HEALTHCARE INDUSTRY

There is a prediction that machine learning will be able to help with disease diagnosis and treatment. Medical systems, by using data will be able to help patients who will be able
to avoid unnecessary tests.4 The use of wearable devices and sensors (eg: Fitbit) utilize data to assess a patient's health in real time.

ML IN FINANCIAL SERVICES

The financial institutions use machine learning technology for two main purposes:
(a) Identify insights in data to help with investments- investors will know when to trade.5
(b) Avoiding frauds- Huge data can be scanned and unusual transactions noted in order to combat fraud.2

ML IN TRANSPORT/LOGISTICS

By 2030, it is predicted that self-driven cars will handle the transportation. On the other hand, taking example of Uber (identifying shortest traffic free routes,
share rides while identifying similar route customers) - all these are done by using machine learning.4

Use of Machine Learning In Different Sectors

ML IN GOVERNMENT

By using machine learning, satellite images are obtained giving granular data about how urbanization is occurring in India. Using big data, which is a part of Machine Learning, the economists have been able to determine trade flows across the states, labour flows as well as the assessment of lifestyle patters of citizens with income declaration (via statistics through banks).6

ML IN RETAIL SECTOR

Retailers use Personalized Product Recommendation using Machine Learning which helps them gain fantastic revenues as these recommendations help increase the conversion rates.4

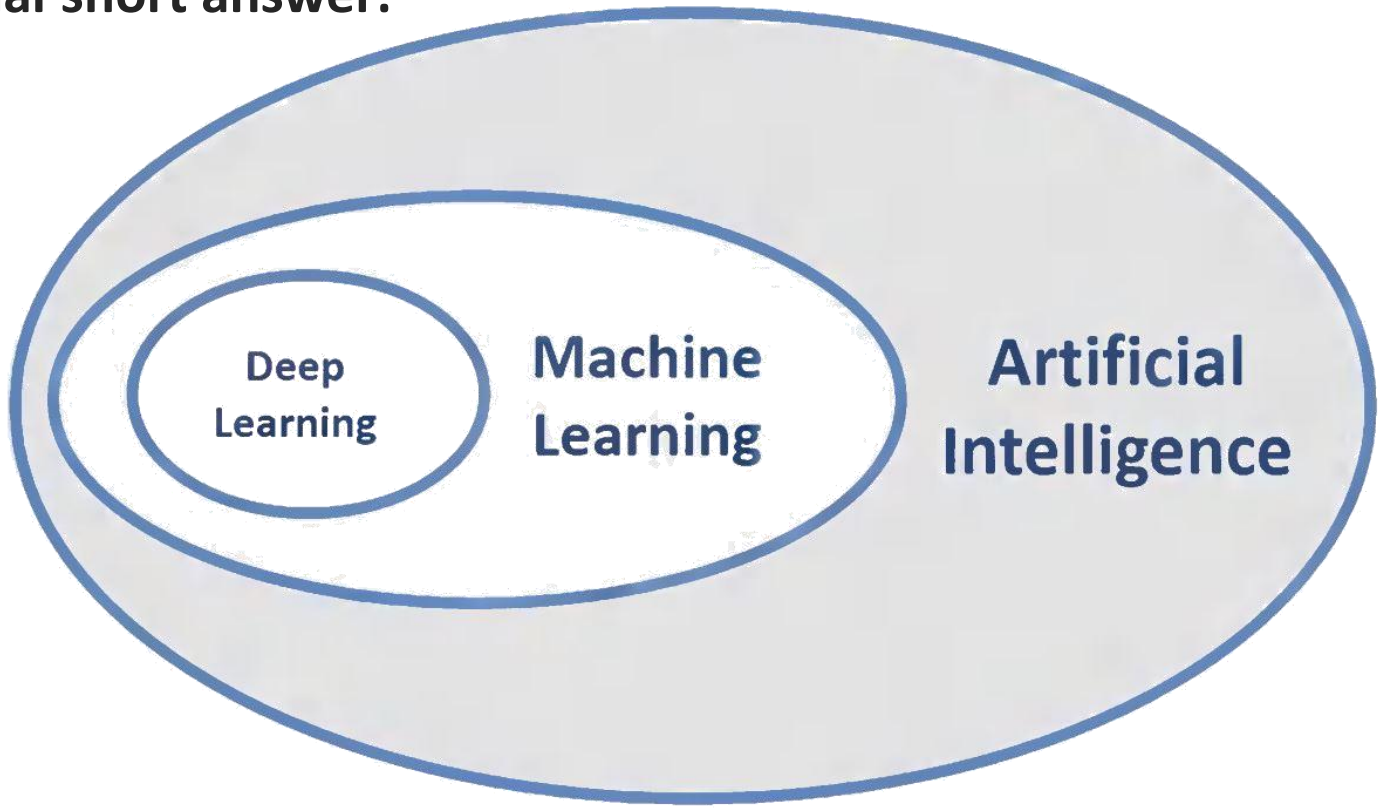
Learning Agents?

"Of particular interest in AI are *Learning agents*, which are capable of changing themselves(Performance measure (utility function)) given training examples or through positive or negative feedback, such that the average utility of their actions grows over time."

- Wolfgang Ertel, "Introduction to Artificial Intelligence," 2nd Edition (2017)

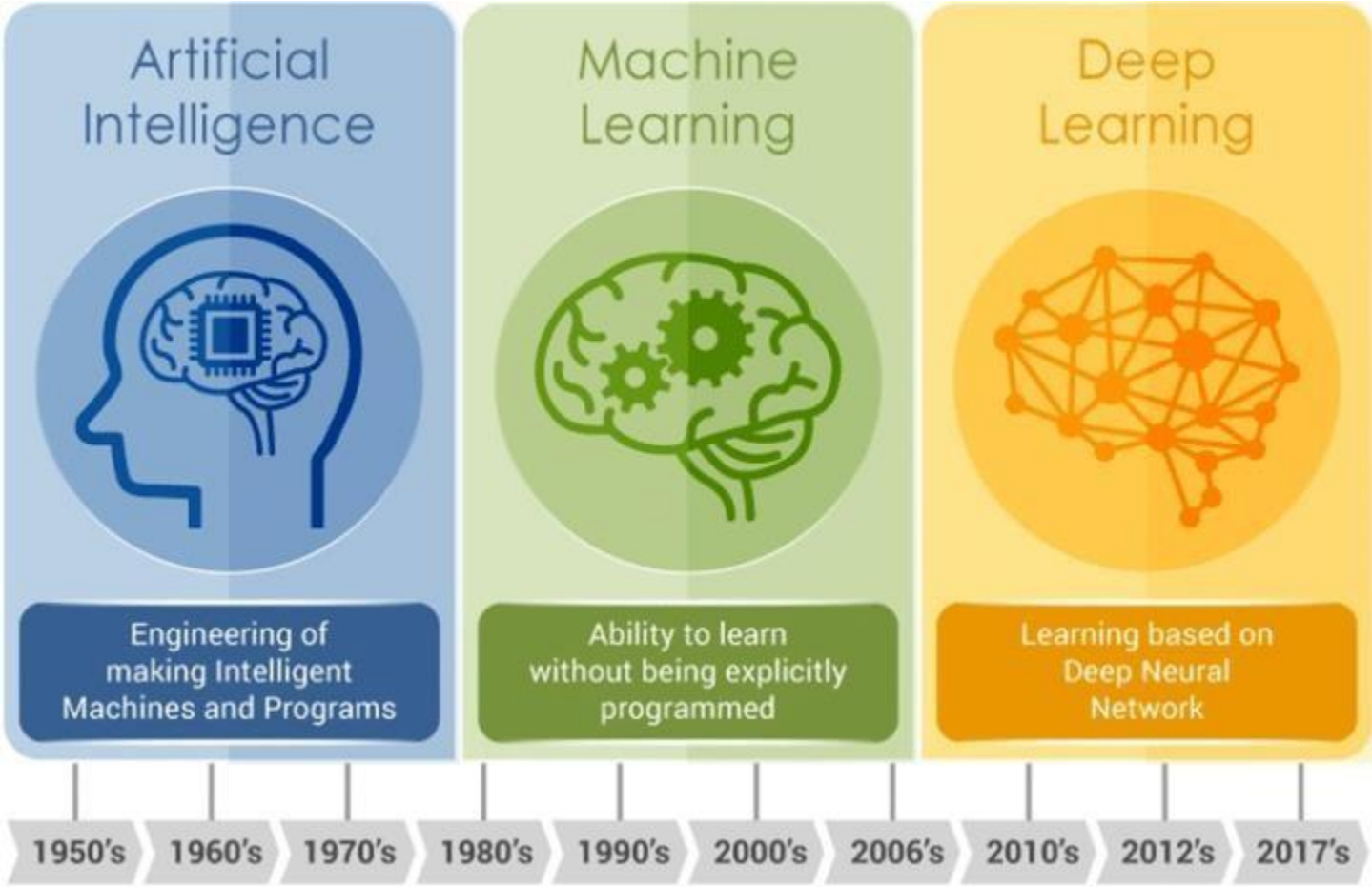
Artificial Intelligence Vs. Machine Learning Vs. Deep Learning ..

A formal short answer:

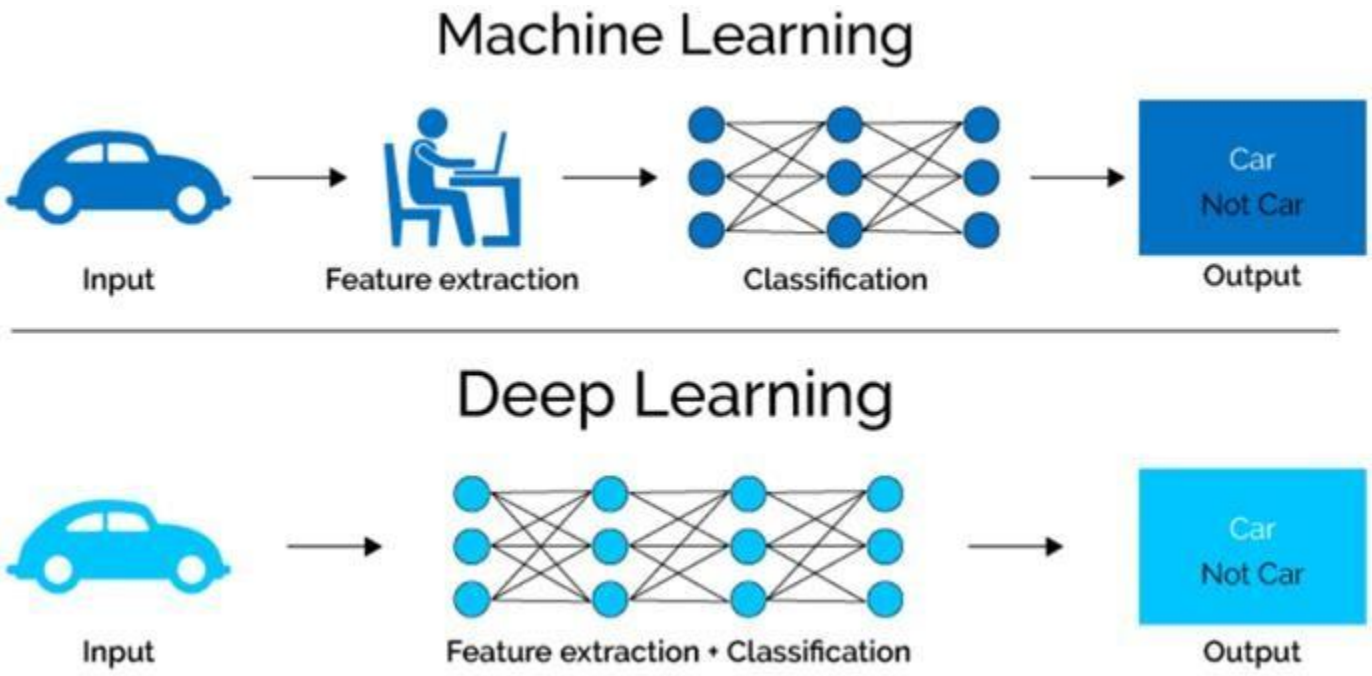


Deep Learning is a part of Machine Learning, which is a part of AI.

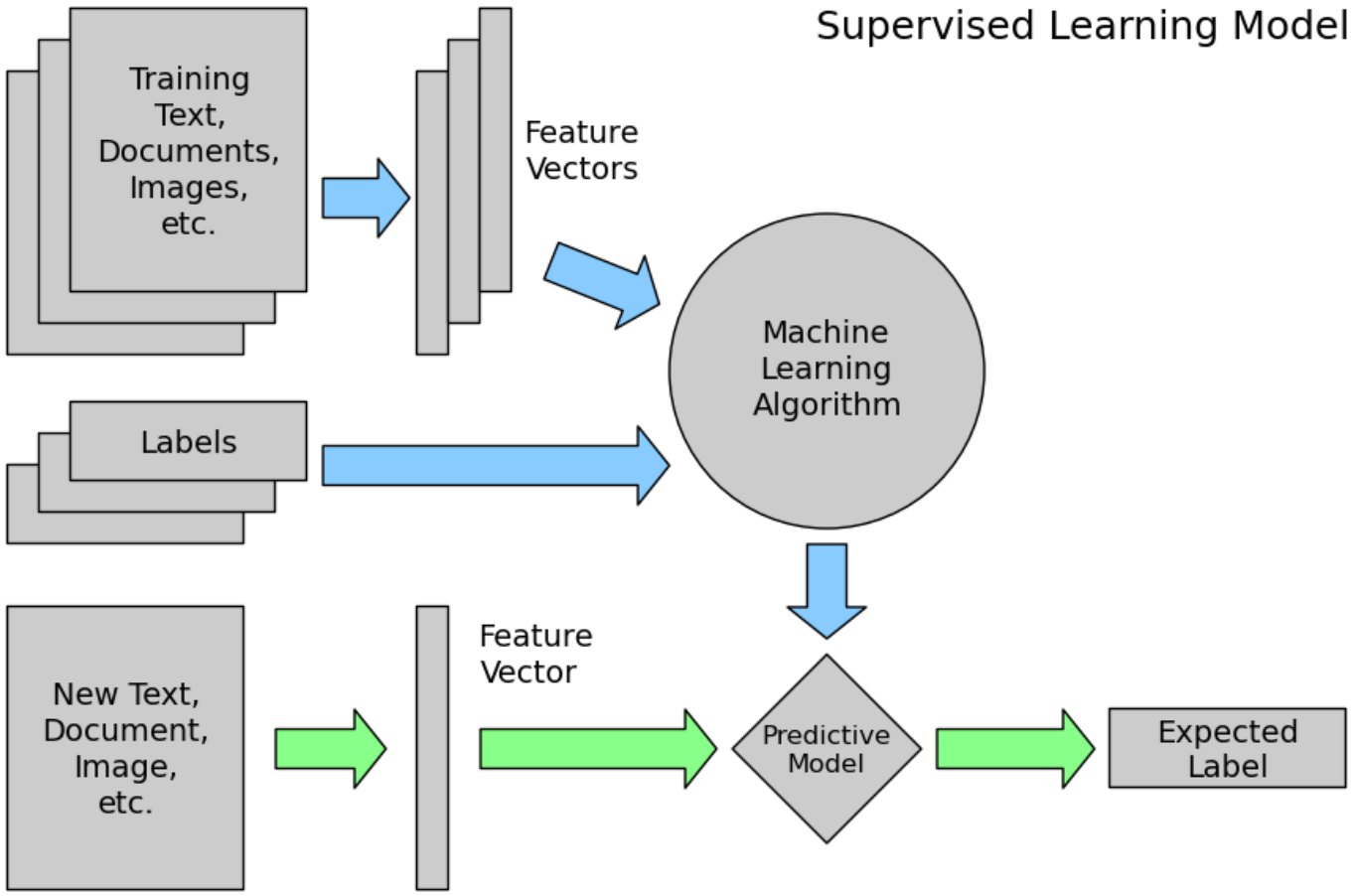
Artificial Intelligence Vs. Machine Learning Vs. Deep Learning ..



Machine Learning Vs. Deep Learning ..



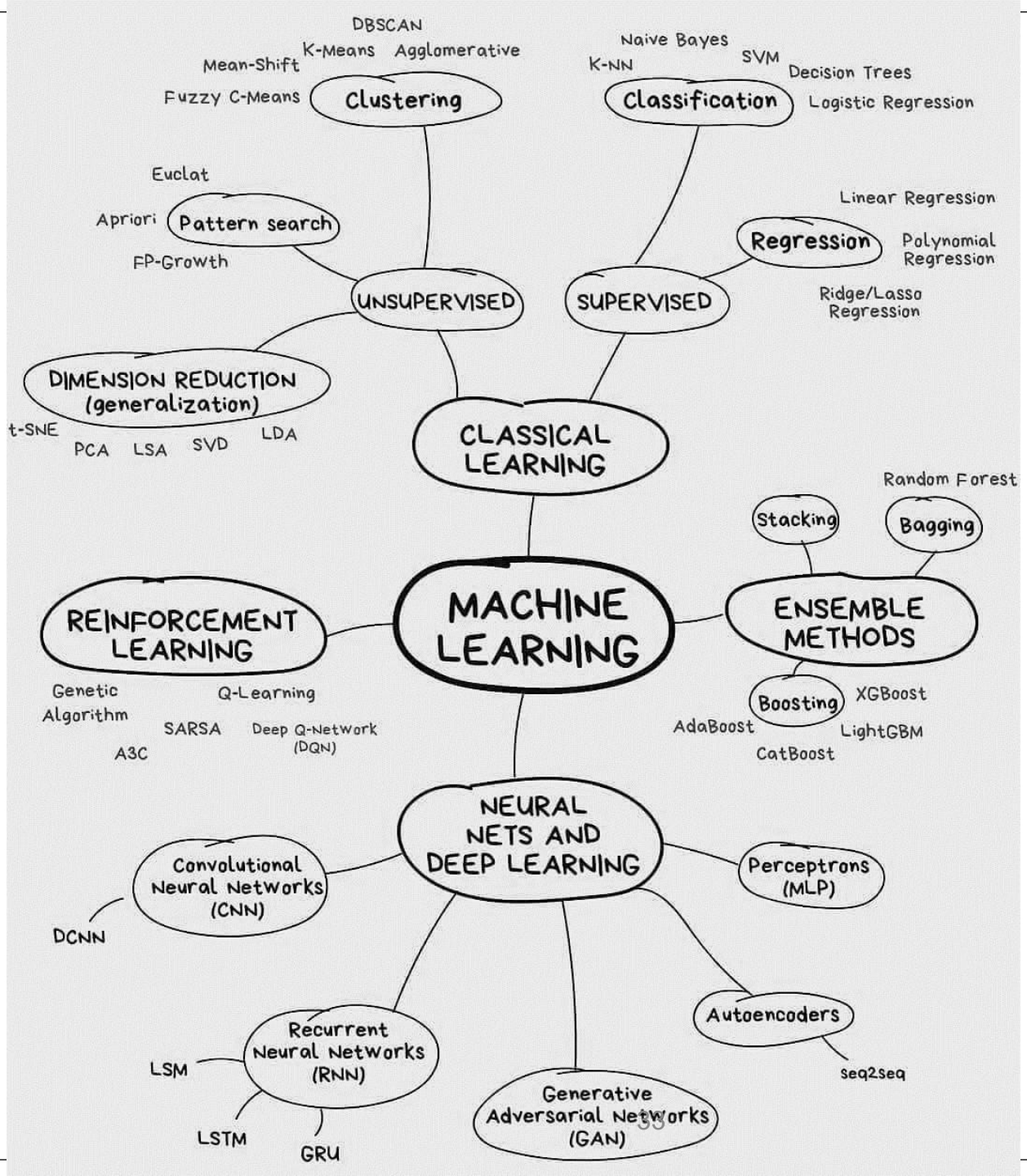
Traditional Machine Learning?

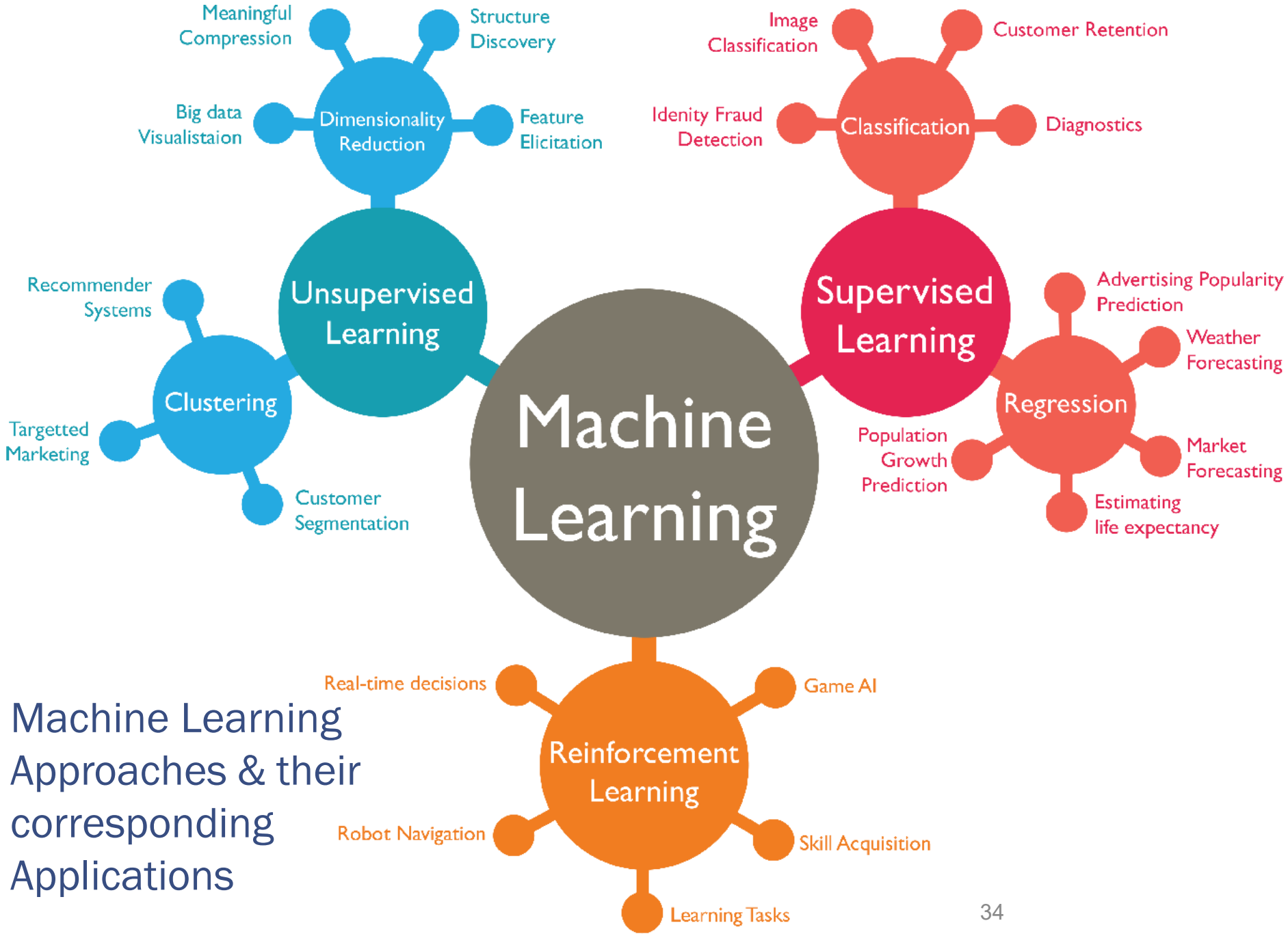


Generic Machine Learning Models

Machine Learning?

{*Artificial Intelligence*}
Machine Learning Map





Machine Learning Approaches & their corresponding Applications

What about Data-Mining?

Machine learning and data mining often employ the same methods and overlap significantly. They can be roughly

- distinguished as follows:

- ***Machine learning focuses on prediction, based on known***
- ***properties learned from the training data.***

- ***Data mining focuses on the discovery of (previously) unknown properties in the data. This is the analysis step of Knowledge Discovery in Databases (Unsupervised learning)***

Much of the confusion between these two research communities comes from the basic assumptions they work with: **in machine learning, performance is usually evaluated with respect to the ability to reproduce known knowledge**, while **in Knowledge Discovery and Data Mining (KDD) the key task is the discovery of previously unknown knowledge**.

Artificial Intelligence Vs. Data-Science

DataScience-Some PossibleDefinitions:

Data Science is the science which uses **computer science**, **statistics** and **machine learning(AI overlap with Data-Science)**, **visualization** and **human- computer interactions** to *collect*, *clean*, *integrate*, *analyse*, *visualize*, *interact* with data to create data products.

Data science = statistics + data processing + machine learning + scientific inquiry + visualization + business analytics + big data + …

Why Data Science?

in **20th Century Innovations ..**
Engineering and Computer Science played key role:

- Cars
- Airplanes
- Power grid
- Television
- Air conditioning and central heating
- Nuclear power
- Digital computers
- The internet

WhyDataScience?

But how about these 20th Century questions?

- Does fertilizer increase crop yields?
- Does Streptomycin cure Tuberculosis?
- Does smoking cause lung-cancer?

WhyDataScience?

What is the difference? .. *Data*

- Does fertilizer increase crop yields?

  Answer: Collect and analyse agricultural experimental **data**.

- Does Streptomycin cure Tuberculosis?

  Answer: Collect and analyse randomized trials **data**.

- Does smoking cause lung--cancer?

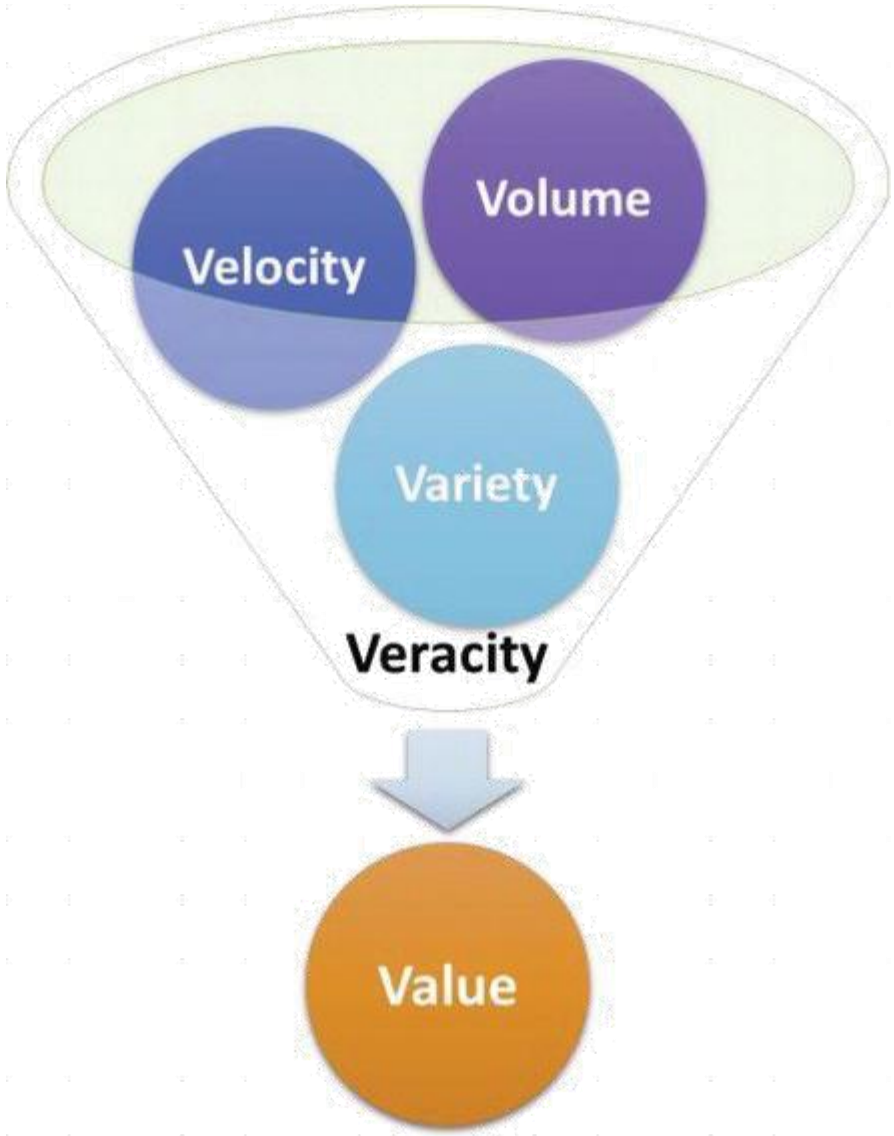
  Answer: Collect and analyse observational studies **data**.

✓ Deductive *versus* empirical ..(Data-Science VS AI(Machine learning))

✓ **Solutions deduced mostly from theory** *versus* **solutions deduced from mostly from data** ..

The Dawn of BigData

To understand the phenomenon that is big data, it is often described using five Vs: *Volume*, *Velocity*, *Variety*, *Veracity*, **and** *Value*.



Recently, Visualization, Virality, & Viscosity were added *(thus, Eight Vs)* ..

The Dawn of BigData

- **Volume** refers to the vast amounts of data generated every second.
- **Velocity** refers to the speed at which new data is generated and the speed at which data moves around.
- **Variety** refers to the different types of data we can now use. In fact, 80% of the world's data is now unstructured, and therefore can't easily be put into tables (think of photos, video sequences or social media updates).
- **Veracity** refers to the messiness or trustworthiness of the data. With many forms of big data, quality and accuracy are less controllable.
- **Value**; It is all well and good having access to big data but unless we can turn it into value it is useless.

https://www.oreilly.com/ideas/practical-applications-of-reinforcement-learning-in-industry
https://www.quora.com/What-are-some-real-world-examples-of-applications-of-machine-learning-in-the-field
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